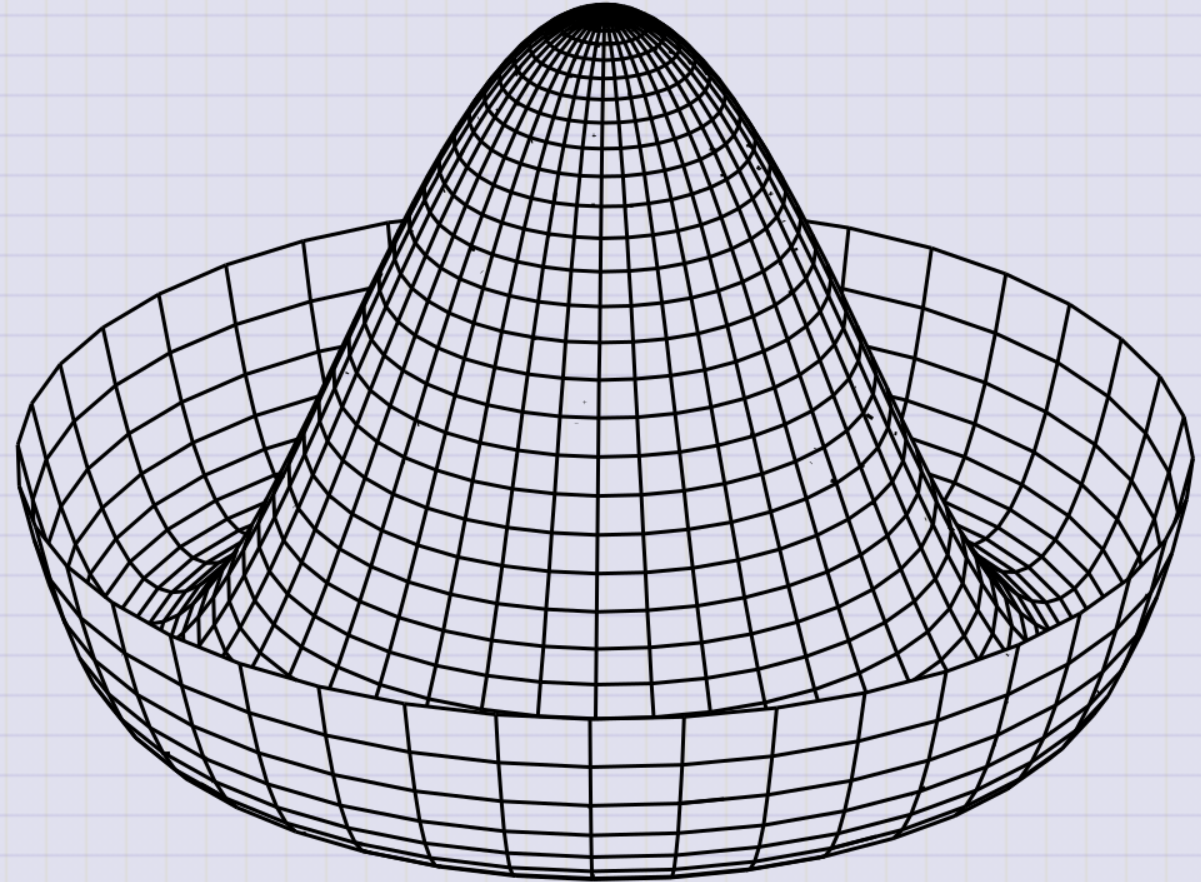


Day 15 - Potential Energy and Stability

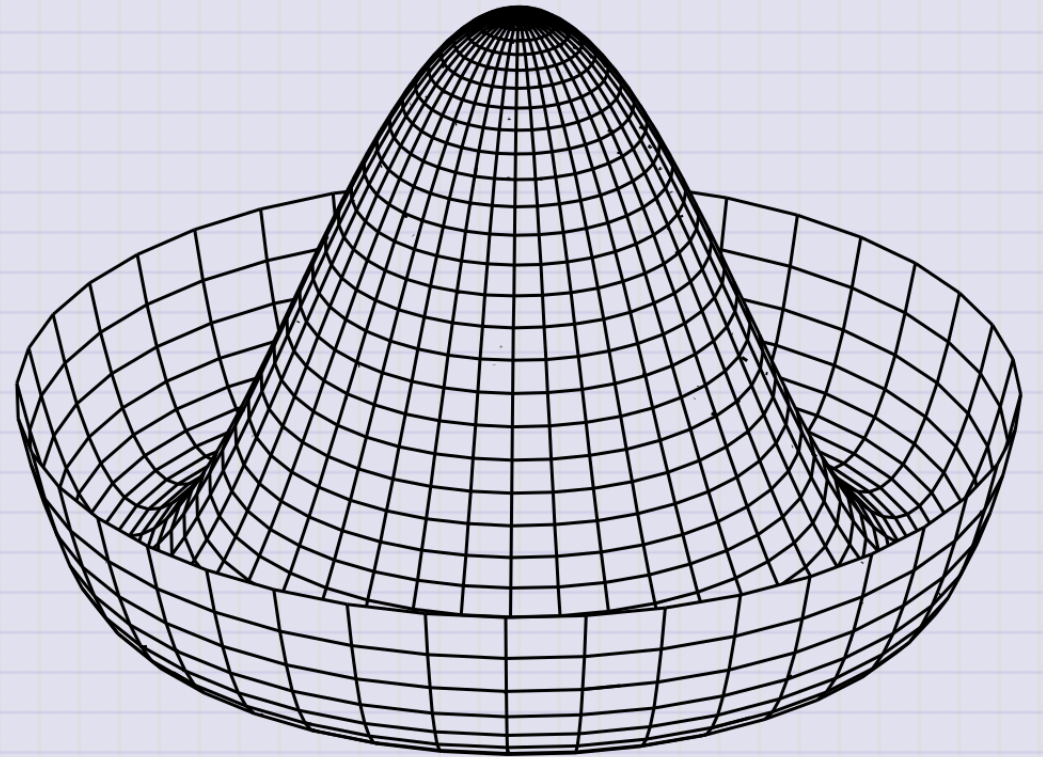
Mexican Hat/Sombrero Potential $\longrightarrow$



Mexican Hat Potential

$$V(\phi) = -5|\phi|^2 + |\phi|^4$$

- Spontaneous Symmetry Breaking (Jeffery Goldstone, 1961)
- Unstable vacuum state at $\phi = 0$
 - Peak of the hat
- Infinite number of stable minima
 - $\phi = \sqrt{5/2}e^{i\phi}$



Announcements

- HW 4 is due today
- Midterm 1 is available today (Due Feb 24th)

Seminars this week

This Week's Goals

- Understand the concept of potential energy
- Determine the equilibrium points of a system using potential energy
- Analyze the stability of equilibrium points
- Define and begin to apply conservation of linear and angular momentum

Reminders: Conservative Forces

- Conservative forces are those with zero curl

$$\nabla \times \vec{F} = 0$$

- The work done by a conservative force is path-independent; on a closed path, the work done is zero

$$\oint \vec{F} \cdot d\vec{r} = 0$$

- The force can be written as the gradient of a scalar potential energy function

$$\vec{F} = -\nabla U$$

